Capstone - BESS*elligence*

Forecasting German Energy Prices for BESS Optimization

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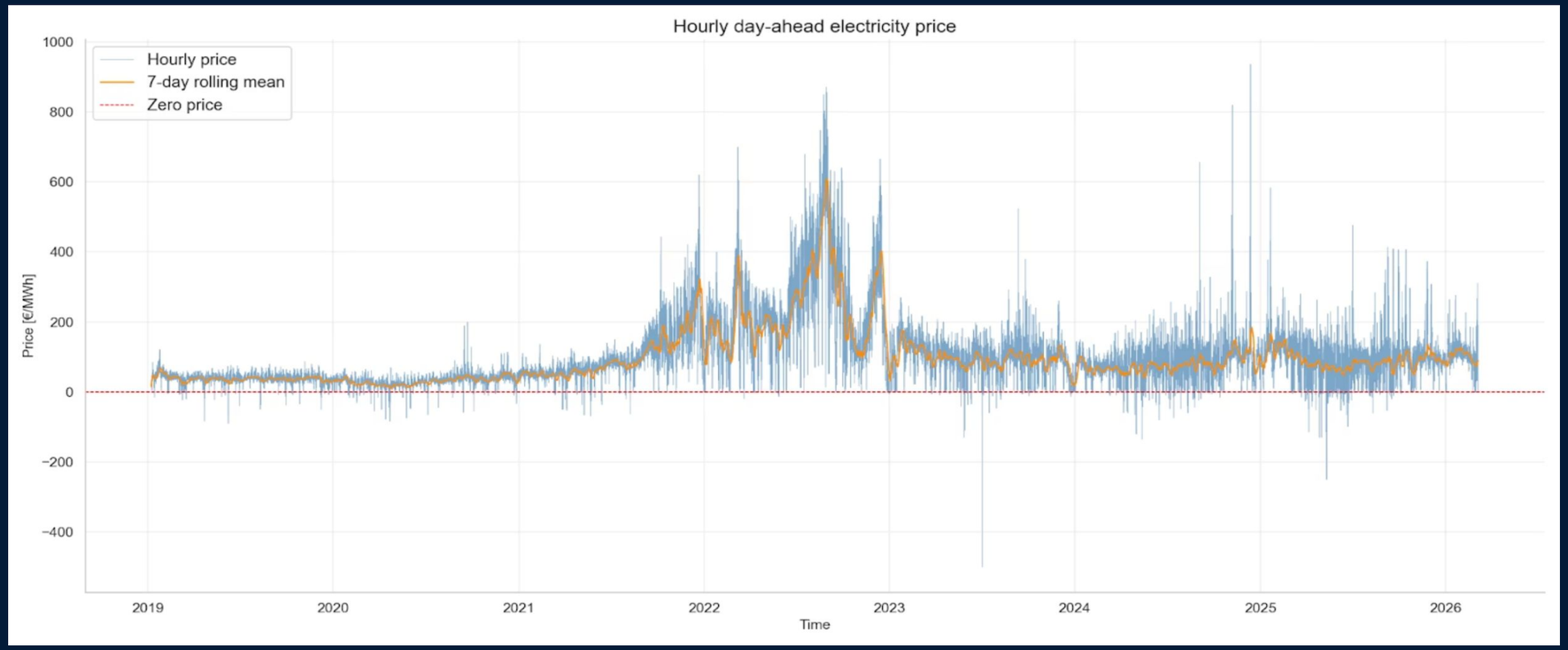
School and Pool for Digital Talent

Germany's Energy Market has changed



Renewable energy share in Germany has surged

Reshaping the entire supply mix

Greater reliance on weather-dependent generation

Introduces **structural price instability**

Day-ahead prices have swung dramatically

From historic highs (>€500/MWh in 2022) to increasingly frequent **negative price hours**

Volatility is no longer an exception

It is the new normal

The Business Challenge

BESS operators face a volatile market where the right dispatch decision can make the difference between profit and loss.

01
What is BESS and why it matters

Battery Energy Storage System (BESS)

Operators manage large-scale batteries that store electricity when supply is abundant and release it when demand is high, helping balance the power system and stabilize prices.

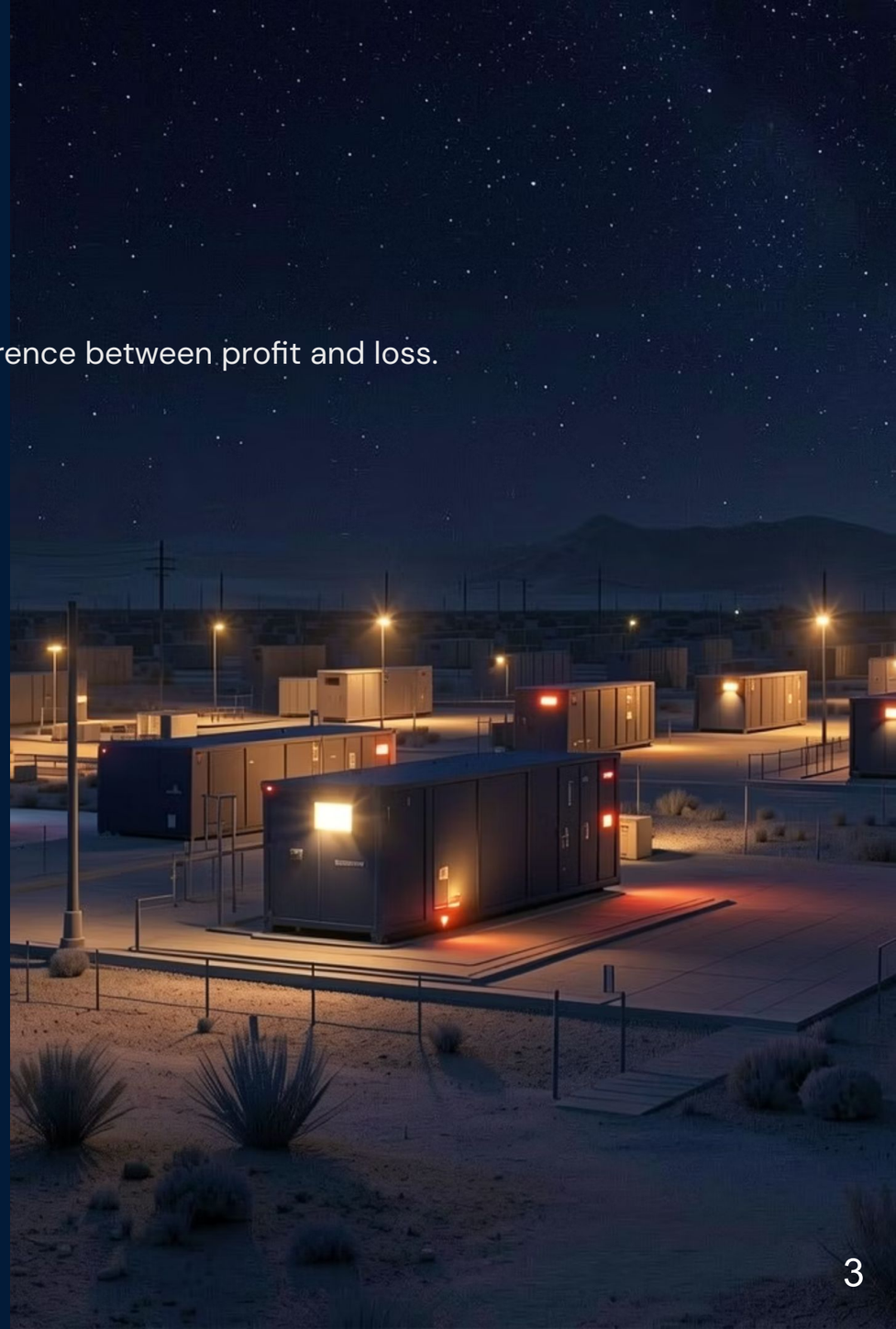
02
The challenge in today's market

In a volatile, renewable-driven market, price spreads are larger than ever — and so is the cost of mistiming.

03
Why timing matters

BESS operators generate revenue by buying cheap and selling expensive — but only if they dispatch at the right hours.

Dispatch decisions must be made day-ahead, based on the price curve for the next 24 hours.



Overview: What BESS*elligence* sets out to solve

Turning volatile electricity prices into better BESS dispatch decisions through forecasting and optimization.

Problem Statement

Wholesale electricity prices are highly volatile, driven by renewable intermittency, demand spikes, and grid congestion. Operators struggle to dispatch BESS without accurate short-term price forecasts.

Our Approach

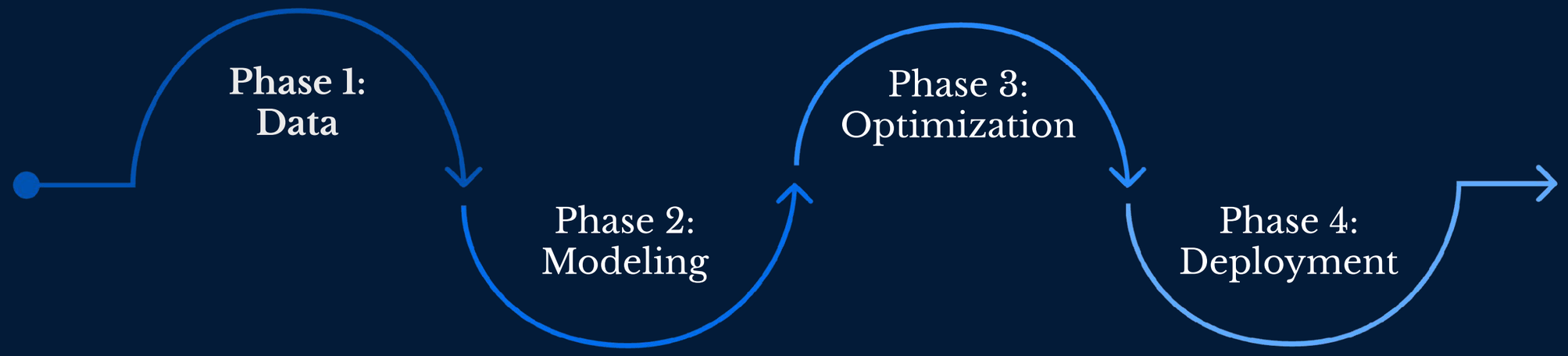
BESS*elligence* combines machine learning-based day-ahead forecasting with a rule-based and reinforcement-learning BESS dispatch optimizer to maximize arbitrage revenue and grid services income.

Deliverables

- End-to-end Python pipeline
- Trained forecasting models
- BESS dispatch optimizer
- Deployed application

Day-Ahead Electricity Market – Auction-based wholesale market where producers, traders, and large consumers buy and sell electricity one day before delivery. A market-clearing algorithm determines prices for each hour of the following day.

Project Roadmap



Critical Milestones

1

Baseline
Complete data pipeline setup, EDA, and baseline model

2

Forecast
ML model results and feature importance analysis

3

Battery optimization
Best model integrated with BESS revenue-max optimization algorithm

4

Streamlit App
Full backtest completed, revenue simulation validated and App connection

Data & Feature Engineering

Raw Data

• Hourly day-ahead price	ENTSO-E
• Hourly load and generation	ENTSO-E
• Coal, gas, CO2 price	Yahoo Finance, ICAP
• Weather forecast	Open-Meteo

Feature Engineering

Only information available at bid submission in the day-ahead market used (no leakage).

Lagged Features

Past prices, volatility, generation, net export (≥ 24-hour lag)

Residual Load

Load minus renewable generation and its change over time (ramp)

Time Features

Hour of day, day of week, month, year, holidays

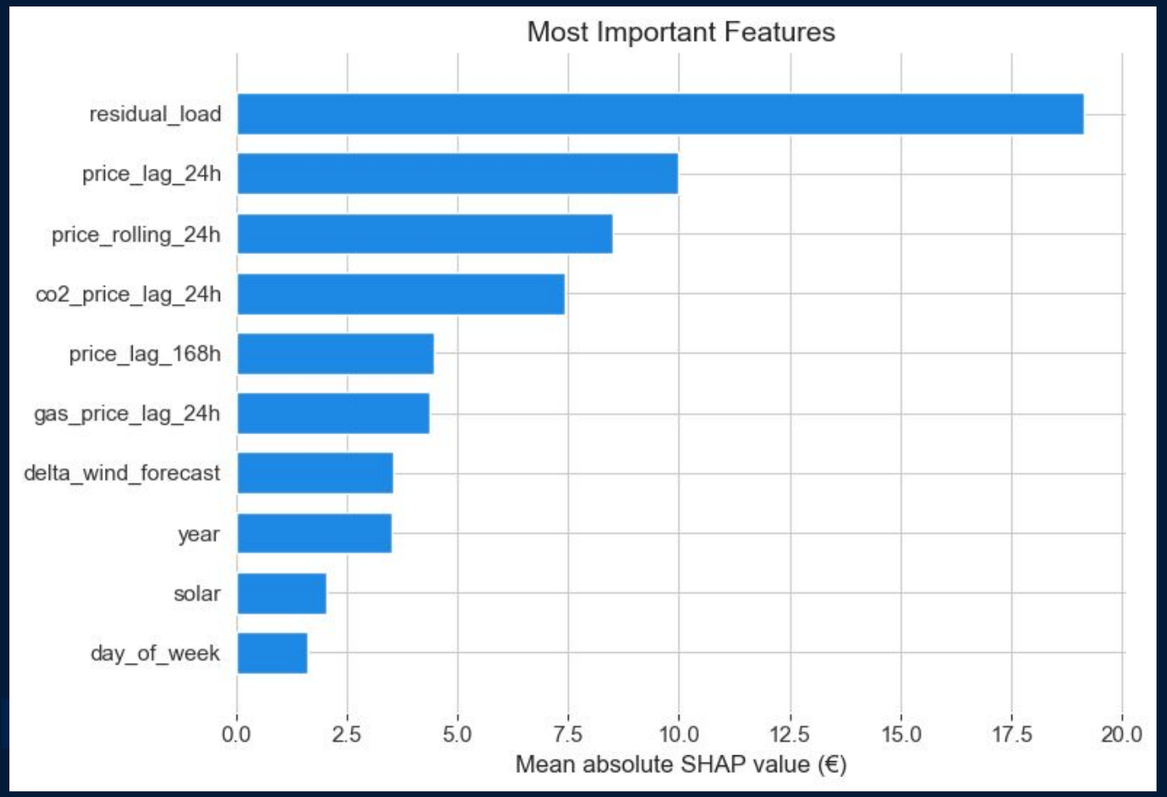
Interactions

Peak hour x Wind generation
Load x Solar generation

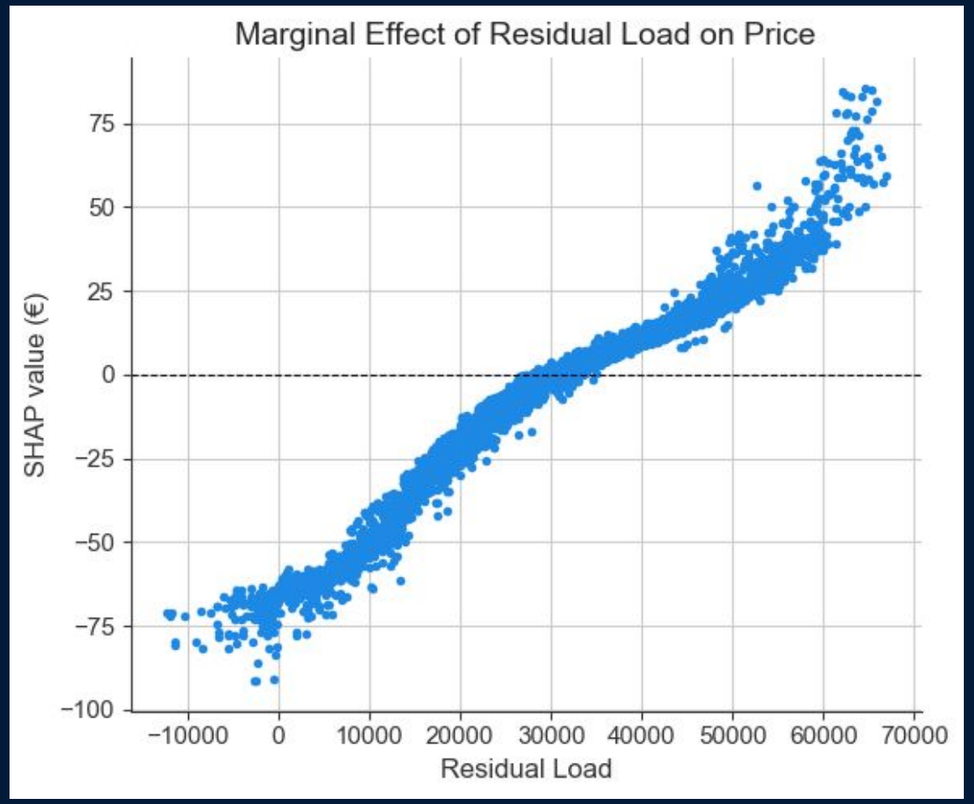
36 features included in the model

What Drives Prices?

Residual load, own price lags and fuel prices as main predictors.



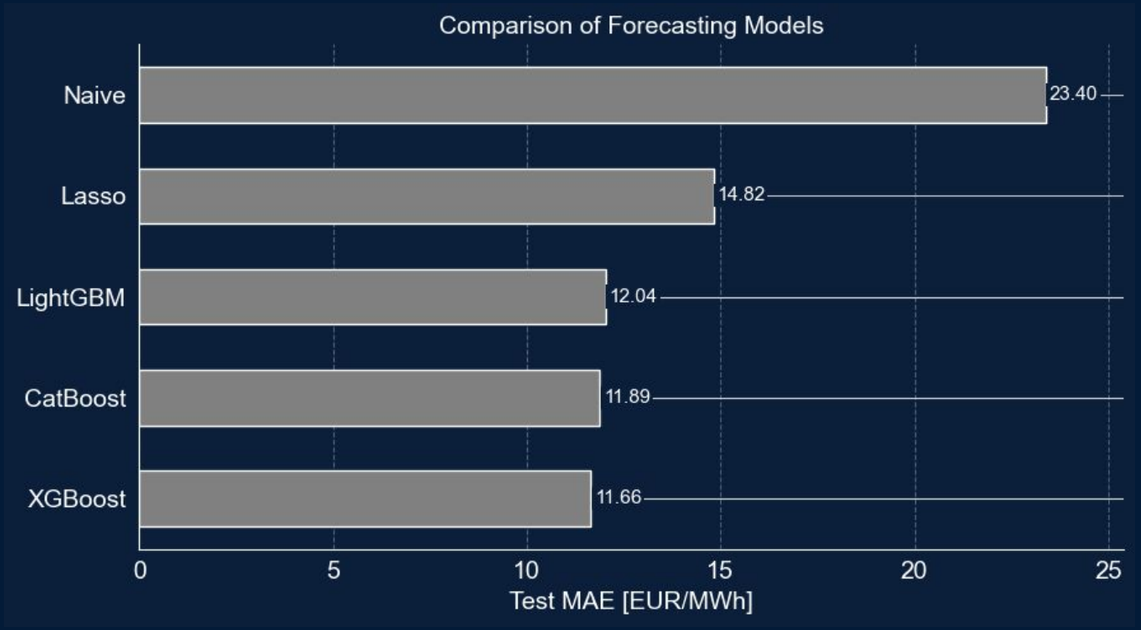
As residual load increases, more expensive generators are dispatched, pushing prices upward.



Residual load is the strongest predictor of day-ahead electricity prices in Germany.

Final Model: LightGBM

After evaluating six models, LightGBM delivered the best predictive performance — selected as the core forecasting engine of BESS*elligence*.



Evaluated Models

Naive
(Baseline)

Lasso Regression
(Baseline)

XGBoost

LightGBM

CatBoost

Naive: uses yesterday's hourly prices; for Monday and Saturday, uses the same day last week.

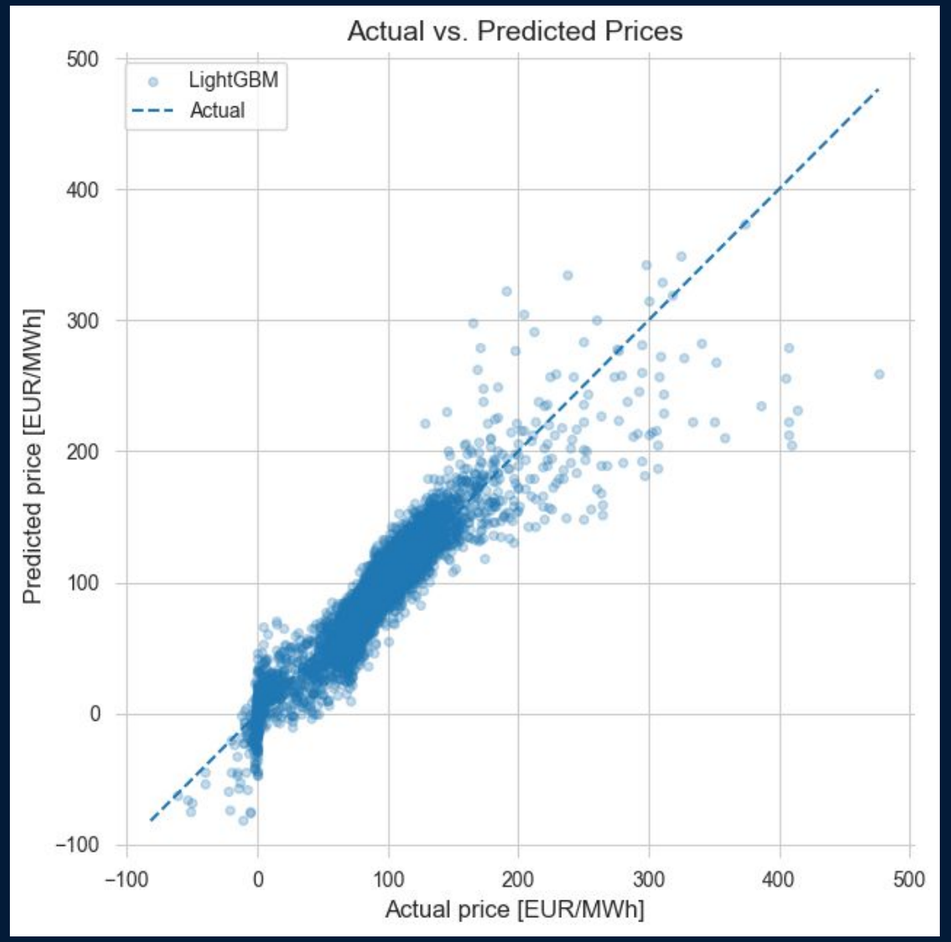
Evaluation Metrics

MAE
Average absolute error across all hourly price forecasts.

DAE
Absolute error in the overall daily average price level.

Model Residuals: LightGBM

The model performs strongly across the full price range — with tighter residuals below €200/MWh and wider spread at extreme price spikes.



- The model captures the overall price trend well, since most points lie close to the 45° line.
- Errors increase as actual prices become higher, shown by the wider spread at the upper end.
- The model tends to underpredict extreme price spikes, because many very high actual prices fall below the diagonal.
- Negative and very low prices are captured less accurately, with larger dispersion in that region.

Key Takeaway

Overall, the model performs well for normal market conditions, but extreme events remain harder to predict.

BESS Optimization Workflow

From Price Forecast to Optimal Dispatch

Day-Ahead Price Forecast
LightGBM predicts the hourly price curve for the next 24 hours.

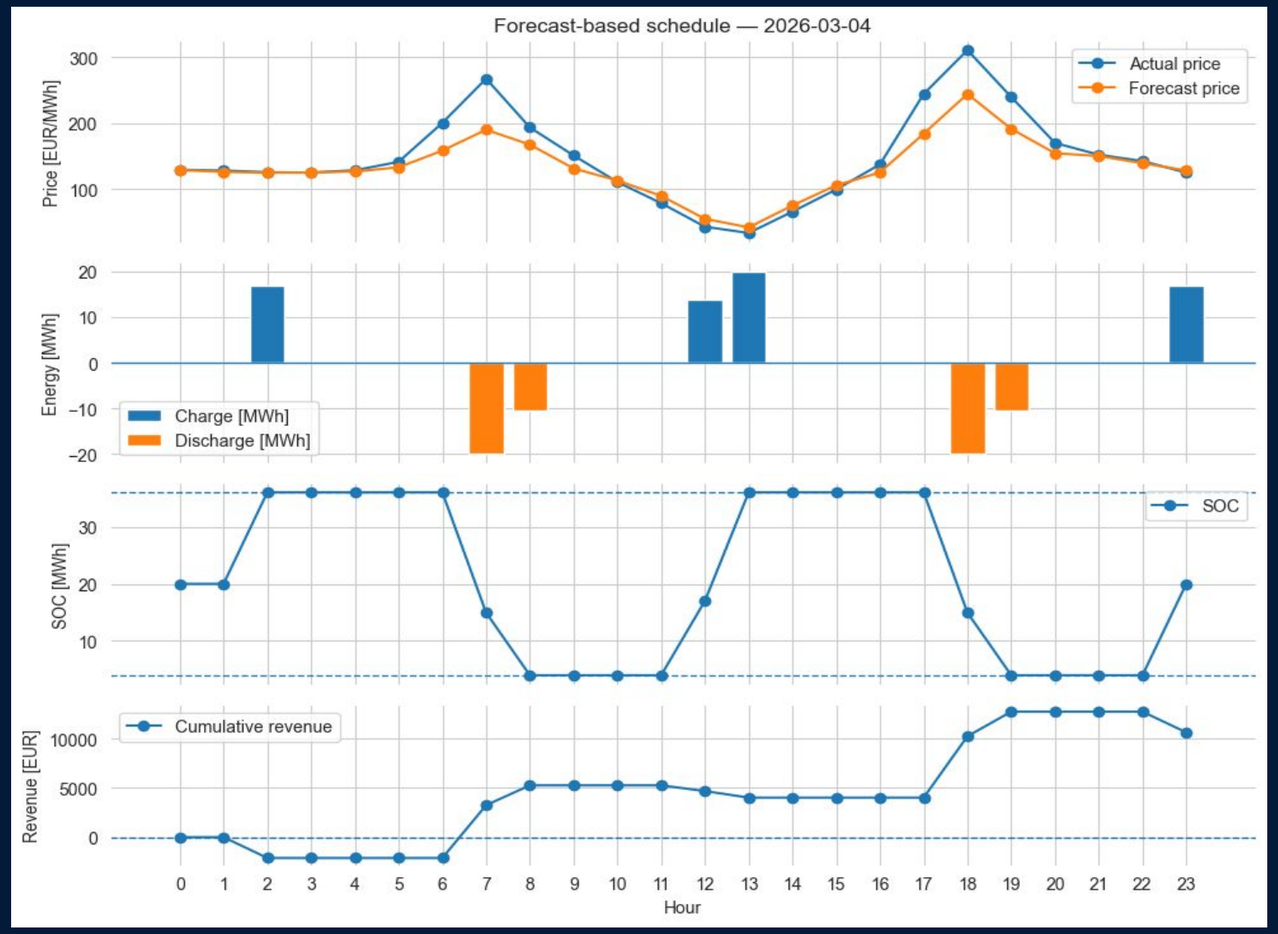
Battery Parameters & Constraints
Capacity, power limits, SoC bounds, and charging/discharge efficiency are defined.

Optimization Problem Formulation
Objective function and operational constraints are set up.

Hourly Dispatch Optimization
The solver determines the optimal charge/discharge schedule for each hour.

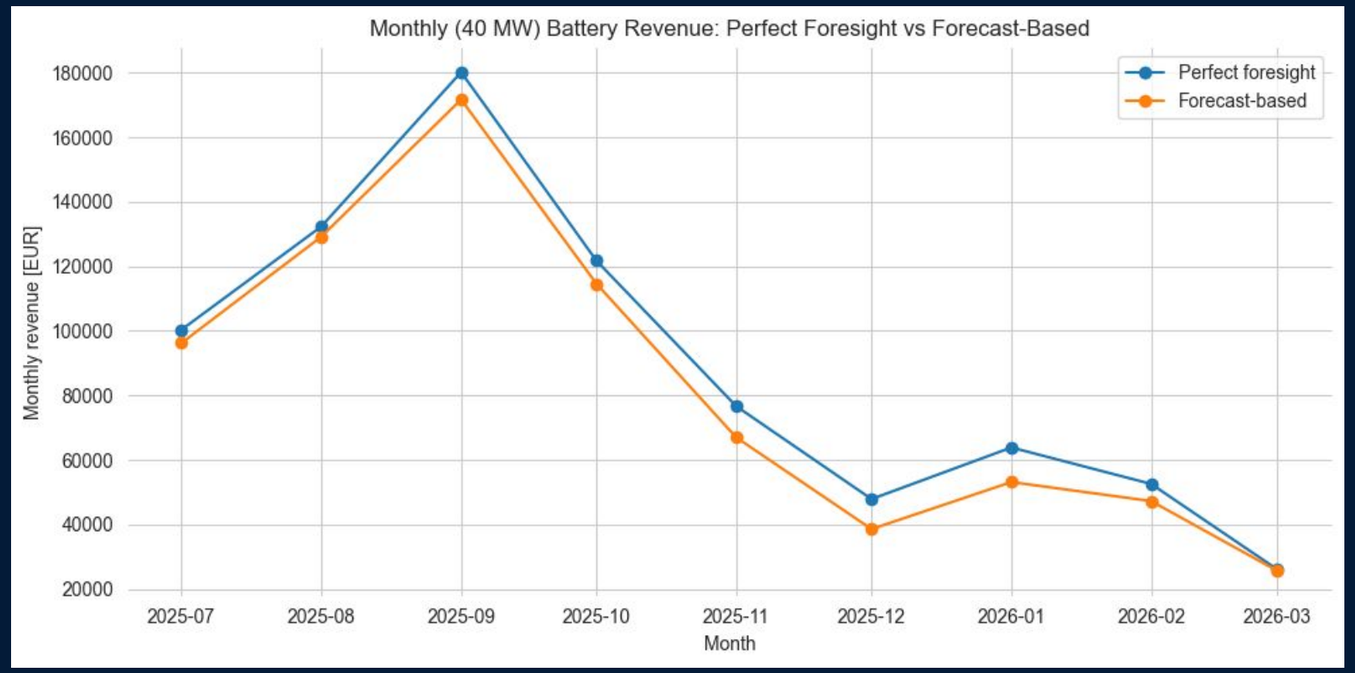
Outputs
Optimal dispatch schedule, SoC profile, and expected revenue.

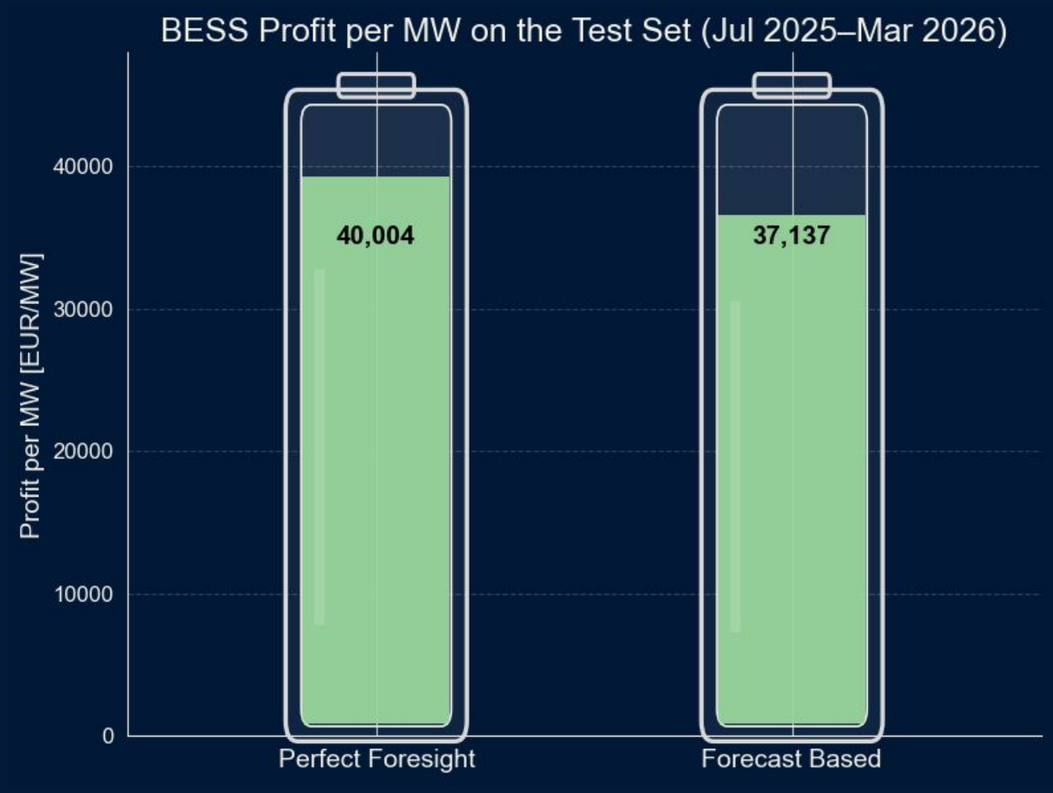
BESS Optimization Workflow



BESS Optimization Output

92.8%





BESS*elligence* — Live Demo

01
Day-Ahead Price Forecasting

02
Battery Charge/Discharge Optimization

03
Revenue Comparison View

Scan the QR code or click the link to explore the live app. See live predictions, watch the optimization algorithm work, and understand the revenue improvement potential.
https://huggingface.co/spaces/Josip007/WattWise

Conclusion

Limitations

- Extreme price events (e.g. 2021–22 crisis spikes) remain difficult to forecast — tail risk is not fully captured
- BESS physical constraints (degradation, efficiency losses, ramp rates) are simplified in the current optimisation
- Scope is limited to day-ahead prices — intraday price movements are not yet exploited

Next Steps

Intraday

Extend the pipeline to capture continuous intraday price dynamics

Balancing Power

Layer in FCR* and aFRR* markets for additional, more stable revenue streams alongside arbitrage

Multi-Market Stacking

Combined, these steps move BESS*elligence* from a single-revenue to a market-stacked ALGO Trader

* FCR = Frequency Containment Reserve ("Primärregelleistung"); aFRR = automatic Frequency Restoration Reserve ("Sekundärregelleistung")

Translate this picture in an avatar

Thank You

Your BESS*elligence Pro*ject Team

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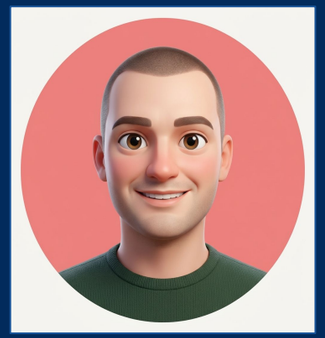

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Scan to open the BESS*elligence* App

https://huggingface.co/spaces/Josip007/WattWise

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